INCH-POUND
MIL-DTL-24643/16G
27 November 2023
SUPERSEDING
MIL-DTL-24643/16F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 1,000 VOLTS, TYPE LSTSGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight with Circuit Integrity)

First	-	Copper conductor, coated or uncoated (see table I for size).
Second	-	Thermoset insulation, extruded or taped (see table I for thickness).
Third	-	An optional glass braid.
Fourth	-	An optional covering (unless required by 3.3.6 of MIL-DTL-24643).
Fifth	-	The three conductors shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm, well-rounded assembly. Standard identification code applied by method 1 or 3, or letter identification code applied by method 5. If identification method 1 is used, outer surface to be printed shall appear black.
Sixth	-	An optional binder.
Seventh	-	Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/16	Type and size	Conductor size		Cable overall diameter max. (inches)	Insulation thickness min. avg. (inch)	Jacket thickness min. avg. (inch)	Conductor resistance (ohms)	Insulation resistance (megohms)	Accelerated service loading (amperes)
		Navy STD	AWG/MCM						
-01UN	LSTSGU-3	--	16 (class B)	0.411	0.018	0.030	4.3	500	--
-02UN	LSTSGU-4	--	14 (class B)	0.449	0.018	0.030	2.68	500	--
-03UN	LSTSGU-9	--	10 (class B)	0.575	0.018	0.040	1.08	500	--
-04UN	LSTSGU-14	--	9 (class B)	0.718	0.018	0.040	0.859	500	90
-05UN	LSTSGU-23	--	7 (class B)	0.812	0.018	0.050	0.543	500	115
-055UN	LSTSGU-30	--	5 (class C)	0.852	0.030	0.050	0.358	225	140
-058UN	LSTSGU-40	--	4 (class C)	0.900	0.030	0.050	0.290	215	160
-06UN	LSTSGU-50	--	3 (class C)	0.969	0.035	0.050	0.210	200	185
-065UN	LSTSGU-60	--	2 (class D)	1.060	0.035	0.050	0.190	190	215
-07UN	LSTSGU-75	--	1 (class C)	1.134	0.035	0.050	0.134	175	250
-08UN	LSTSGU-100	--	0 (class D)	1.266	0.035	0.060	0.106	160	295
-085UN	LSTSGU-125	--	00 (class D)	1.408	0.040	0.060	0.0888	150	340
-09UN	LSTSGU-150	--	000 (class D)	1.515	0.040	0.060	0.0674	135	400
-10UN	LSTSGU-200	--	0000 (class D)	1.669	0.050	0.060	0.053	125	460
-105UN	LSTSGU-250	--	250 MCM (class C)	1.794	0.050	0.060	0.0453	120	535
-11UN	LSTSGU-300	--	300 MCM (class D)	1.957	0.050	0.075	0.0377	110	590
-115UN	LSTSGU-350	--	350 MCM (class D)	2.073	0.050	0.075	0.0321	105	665
-12UN	LSTSGU-400	400 (127)	--	2.203	0.050	0.075	0.0273	100	750

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	
Sizes 3 through 9	3,000
Sizes 14 and larger	5,000
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	See table I
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Abrasion resistance – scrapes minimum (sizes 40 and smaller) (extruded insulation only)	250
Crack resistance (sizes 3 through 100) (extruded insulation only)	No damage

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation (extruded insulation only)	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C]) (see MIL-DTL-24643 for limits of drip allowance)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160

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Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation, method 1 only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure

Group D:

Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Cold working (-4±3.6 °F [-20±2 °C])	No damage
Gas flame (circuit integrity) (3 hours)	No failure
Accelerated service (sizes 14 and larger) (see table I for load current)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50

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Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption (extruded insulation only)	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LSTSGU	3	2,500
LSTSGU	4 through 40	2,000
LSTSGU	50 and 60	1,500
LSTSGU	75 through 150	1,000
LSTSGU	200 through 400	800

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-032)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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